TCS — Aptitude Test 1

Questions 1–25 of 100 · 25 questions

Question 1. If given equation is 137+276 = 435, how much is 731+672=?

Solution:
We know that 137+276 ≠435 in normal addition.
If you see the second equation, it is the exact reverse of first equation.
The answer for second equation must be exact reverse of answer of first equation.
The answer is
731 + 672 = 534

1.67
8.33
:

The ratio between the two varieties is
1 : 5.

Question 2. If the ratio of two numbers is 3:4 and LCM of the number is 180 then what is the number.

Solution:
Formula:
Product of two numbers = Product of (LCM x HCF)
Assume that the two numbers are
3x and 4x.
HCF = x
■LCM = 180
From the formula, we get
(3x)(4x) = 180x
■12x2 = 180x
x = 15.
Sub x = 15 in 3x and 4x
3x = 45; 4x = 60
The numbers are 45 and 60

Question 3. (1/3)rd of a number is 6 more than (1/6)th of that number. Find the number.

Solution:
Assume that the number is x.
(1/3)x – (1/6)x = 6
(1/6)x = 6
x = 36.

3.5

30
+

36

Time taken to fill the tank by pipe 2 with the clog = 36 x (10/9) = 40 minutes.
The fraction of tank filled by pipe 2 in two minutes, when the clog is there is,
= 1/40
Total fraction of tank filled by pipe 2 is:

3.5

36
+

40

Combined work:

N

3.5

30
+
N

3.5

36
+
+
= 1

36

40

By solving the above equation, we get
19N + 77 = 360
19N = 283; N ■15 minutes

Solution:
Let S1 be the speed of Ram and S2 be the speed of Shakil.
Time taken by Ram in first case
= 2000/S1
Time taken by Shakil in first case
= 1800/S2
From the question,
(2000/S1) = (1800/S2) – 1 ---(2)
Time taken by Ram in second case
= 1000/S1
Time taken by Shakil in second case

Question 4. When Rs.250 added to 1/4th of a given amount of money it makes it smaller than 1/3rd of the given amount of money by Rs 100. What is the given amount of money?

Solution:
Assume that the given amount of money is Rs. X
(1/4)X + 250 – (1/3)X = 100
By solving, we get X = 1800

Question 5. The sum of the digits of a three digit number is 17, and the sum of the squares of its digits is 109. If we subtract 495 from the number, we shall get a number consisting of the same digits written in the reverse order. Find the number.

a. 773

b. 683

c. 944

d. 863

Solution:
Best method to solve the question is to go with the options.
Sum of the squares of the digits is 109.
Only possibility is 82 + 62 + 32 = 109
The number may be 863 or 683 according to options.
863 – 495 = 368. This the exact reverse.
The required number = 863

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Question 6. Taxi fare is equal to 15 Rs/km & Train fare is equal to 21 Rs/km. If total distance traveled is equal to 450 km and total amount charged is equal to Rs. 8320. Then distance traveled by train is?

Solution:
Assume that the distance travelled by train is X.
The distance traveled by bus will be,
450 – X
Total fare = Train fare + Bus fare
8320 = 21(X) + 15(450 – X)
8320 = 6X + 6750
X = 261.67 km

Question 7. Anand packs 304 marbles into packets of 9 or 11 so that no marble is left. Anand wants to maximize the number of bags with 9 marbles. How many bags does he need if there should be at least one bag with 11 marbles?

Solution:
Let X be number of bags with 11 marbles
(304 – 11X) should be a multiple of 9.
Substitute X = 1, 2, 3, … till the above equation is satisfied.
By solving we get, X = 8
8 bags with 11 marbles and 24 bags with 9 marbles

Question 8. We have an equal arms two pan balance and need to weigh objects with integral weights in the range 1 to 40 kilo grams. We have a set of standard weights and can place the weights in any pan (i.e.) some weights can be in a pan with objects and some weights can be in the other pan. The minimum number of standard weights required is:

Question 9. Divide 50 into two parts so that sum of the reciprocal is 1/12 ?

Solution:
Assume that the two parts of 50 are A and B.
So, A+B = 50
Given, (1/A) + (1/B) = (1/12)
(1/A) + (1/B) = [(A+B)/AB] = (1/12)
50/AB = 1/12
AB = 600
A + B = 50; AB = 600, by solving this, we get
A = 30 and B = 20

Question 10. M men agree to purchase a gift for Rs. D. If three men drop out how much more will each have to contribute towards the purchase of the gift ?

Solution:
Initial Share
= D/M
Final Share
= D/(M-3)
Extra amount for each person
= [D/(M-3)] – (D/M)

= 3D/(M2 – 3M)

Question 11. Three variables x,y,z have sum of 30. All three of them are non- negative integers. If any two variables don't have same value and exactly one variable has a value less than or equal to three, then find the number of possible solution for variables.

Solution:
If x = 1, then y + z = 29 and y or z is greater than 3.
If x = 1, y = 4 then z = 25.
There are 22 different values for y(from 4 to 25) and z value will change accordingly.
When x = 1 , the possibilities are 22
Similarly when y = 1, there are 22 ways and when z = 1, there are 22 ways.
Total when one variable is 1 is = 66
If x = 2, then y + z = 28.
If x = 2 and y = 4, then z = 24.
There are 20 ways for y and z in which y = 14 and z = 14 will not be considered.
Total ways when one variable is 2 is:
19 x 3 = 57.
If x = 3, then y + z = 27.
If x = 3, y = 4, then z = 23
Total ways for y and z = 19.
Total ways when one variable is 3 is:
3 x 19 = 57.
Total possibilities = 66 + 57 + 57 = 180

Solution:
If both the pans can be used for weighing, the weights required are 30, 31, 32, 33, 34 …
The weights required are 1, 3, 9, 27.
For example, if we have to weigh 23 kg, Keep 27kg stone in one pan and 1kg stone, 3kg stone and the material in another pan. The material will weigh 23 kg.
The

number
of
different
weights
required is 4

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Question 12. If 28a+30b+31c = 365 then value of a + b + c = ?

Solution:
With only
one equation
with three
unknown variables, it is impossible to
find the exact solution.
This question is a puzzle.
There are 28 days in February,
30 days in 4 months of a year and
31 days in 7 months of a year.
So, a = 1;
b = 4 and
c = 7
a + b + c = 12
Or we know that obviously there are 12
months in a year.
No need to add, the answer is 12.

Question 13. When a number is divided by 357 it leaves a remainder 5. If the same number is divided by 17, what is the remainder?

Solution:
357 is a multiple of 17.
So the number which was divided by
357, when it is divided by 17, it leaves
the same remainder 5.

Question 14. What is the remainder when (34^31^301) is divided by 9?

Solution:
((3431)301)/9 = ((17x2)31)301)/9
= ((1731)301) x ((231)301)/9

= (179331 x 29331)/9
Reminder of (17/9) = 8 or 8 – 9 = -1
So,
(-19331 x 29331)/9
= (-1 x 29331)/9
Reminder of 21/9 = 2
Reminder of 22/9 = 4
Reminder of 23/9 = 8
Reminder of 24/9 = 7
Reminder of 25/9 = 5
Reminder of 26/9 = 1
Reminder of 27/9 = 2
Reminder of 28/9 = 4
Here powers of two has reminder cycle 6
when divided by 9.
Reminder of 9331/6 = 1
So, reminder of 29331/9 = 2
Reminder of (-1 x 29331)/9 = (-1 x 2)/9
= -2 or (9 – 2) = 7

Question 15. What is the remainder when 617+ 176is divided by 7?

Solution:
Reminder of 61/7 = 6
Reminder of 62/7 = 1
Reminder of 63/7 = 6
Reminder of 64/7 = 1
So, reminder of 617/6 = 6
Reminder of (17)6 /7
= reminder of(14 + 3)6/7
= reminder of 36/9 = 729/7 = 1
Reminder of (6 + 1)/7 = 0

Question 16. What is the remainder when 46! is divided by 47?

Solution:
Reminder of 2!/3 = 2
Reminder of 4!/5 = 4
Reminder of 6!/7 = 6
When a factorial is divided by
the
following number and if the following
number is a prime number the reminder
will be the factorial number itself.
Since 47 is a prime number,
Reminder of 46!/47will be 46.

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Question 17. What is the remainder when

1!+2!+3!+4!+5!+.....+50! is divided by 5!

Solution:
Starting from 5! onwards, all the other factorials are a multiple of 5! i.e. 120.
So, we have to find out the reminder only for sum of factorials from 1 to 4.
1! + 2! + 3! + 4! = 33
Reminder of 33/120 = 33.
Answer = 33.

Question 18. How many trailing zeros are there in 100! After the last significant digit?

Solution:
The formula to find out the number of trailing zeros in n! is:
(n/5) + (n/52) + (n/53) + …
Here n = 100, which is less than 53

So the formula is:
(n/5) + (n/52) = (100/5) + (100/25)
= 20 + 4 = 24
Alternate method:
For every multiple of 10 from 10 to 100 we will have 11 zeros.
For every even number multiplied with a number with unit place 5, there will be 10 zeros.
Example: 32 x 35 will leave a zero.
25 is 5 x 5
50 is 5 x 10
75 is 3 x 5 x 5.
For these extra 3 5's we will get 3 zeros
Total = 11 + 10 + 3 = 24

Question 19. What is the highest power of 7 that will exactly divide 56!?

Solution:
Assume that the number is X.
X is a multiple of 5.
So, X = 5Y.
---(1)
When X is divided by 5, the quotient is Y and reminder is zero.
Y = 3Z + 2,
---(2)
When Y is divided by 3, the quotient is Z and reminder is 2.
Z = 2A + 1
---(3)
When Z is divided by 2, the quotient is A and the reminder is 1.
Substitute (3) in (2)
Y = 3(2A + 1) + 2 = 6A + 5 ---(4)

Substitute (4) in (1)
X = 5(6A + 5) = 30A + 25.
For A = 0, X = 25.
Since LCM of 5, 3 and 2 is 30, we can assume that the number is 25 itself.
Reminder of 25/2 = 1; Quotient = 12
Reminder of 12/3 = 0; Quotient = 4
Reminder of 4/5 = 4
The required answer is (1, 0, 4).

Solution:
The product of numbers from 1 to 56 i.e. 56! has the following multiples of 7.
7, 14, 21, 28, 35, 42, 49, 56.
Among these multiples 49 can be written as 7x 7.
So there are 9 number of 7's in 56!.
Highest power of 7 that can divide 56! Is 9.

Question 20. Find the greatest number that will divide 43,91,183 so as to leave the same remainder in each case.

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Solution:
Take the difference between the successive numbers.
91 – 43 = 48.
183 – 91 = 92.
The answer for the question is the HCF of the differences.
HCF(48, 92) = 4.
So, when the numbers 43, 91 and 183 are divided by 4, you will get the same reminders.

Question 21. The numbers 272738 and 232342, when divided by n, a 2 digit number leaves a remainder 13 and 17 respectively. Find the sum of digits of n?

Solution:
Write all the numbers one by one, to get 21 "8's" in the unit place, 20 "8's" in the tens place and 19 "8's" in the hundreds

place.
Sum of all 8's in units place
= 21 x 8 = 168
8 will be the unit place of the result.
16 is carried over to tens place values.
Sum of all numbers in tens place plus the carry over
= 20 x 8 + 16
= 160 + 16 = 176
6 will be the tens place of the result and
17 is carried over to the hundreds place.
Sum of all numbers in hundreds place plus the carry over
= 19 x 8 + 17 = 169.
9 will be the hundreds place of the result
The last three digits of the sum is
968.

Solution:
(272738 – 13) = 272725
is exactly divisible by n.
(232342 – 17) = 232325
is exactly divisible by n.
The last two digits of the numbers after removing the reminders is 25.
So both numbers are exactly divisible by 25.
n = 25. Sum of the digits
= 2 + 5 = 7.

Question 22. When a number is successively divided by 5, 3 and 2 leaves a remainder of 0, 2 and 1 respectively. What are the remainders when the same number is divided by 2, 3 and 5 successively?

Question 23. 8+88+888+.....+8888.....8888. There are 21 "8" digits in the last term of the series. Find the last three digits of the sum.

Question 24. What is the number of zeros at end of number 28! + 29!?

Solution:
28! + 29!
= 28! + (29 x 28!)
= 28! (1 + 29)
= 28! (30)
Number of zeros in 28!
= Quotient of (28/5) + (28/25)
= 5 + 1 = 6
The number of zeros at the end of 28! is 6.
When 28! is multiplied with 30, there

will be 7 zeros at the end.

Question 25. Find the last two digits of the expression (1141)^3843+(1961)^4181

Solution:
1 raised to any number will give 1.
Unit digit of 11413843 = 1.
Tens digit of 11413843

= tens digit of base value x unit digit of power value
= 4 x 3 = 12.
Tens digit of 11413843 = 2
Last two digits of 11413843 = 21
Unit digit of 19614181 = 1
Tens digit of 19614181 = 6 x 1 = 6
The last two digits of 19614181 = 61
The last two digits of
11413843 + 19614181

= 21 + 61 = 82.